

# RetailCo

## Data Pipeline — Stage 8 Report

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End-to-End Modern Data Engineering Pipeline

Airflow • PostgreSQL • dbt • dlt • Docker • Kimball Dimensional Modelling

**June 2026**

# 1. Executive Summary

This report documents the design, implementation, and validation of the RetailCo end-to-end data pipeline built as part of Stage 8 of the Data Operations programme. The pipeline extracts transactional data from a live ERP REST API, loads it into a PostgreSQL data lake, transforms it through a multi-layer dbt project following Kimball dimensional modelling principles, and orchestrates all steps via Apache Airflow — all running inside a fully containerised Docker environment.

|                                                                |                                                                                     |                                                              |                                                                          |
|----------------------------------------------------------------|-------------------------------------------------------------------------------------|--------------------------------------------------------------|--------------------------------------------------------------------------|
| <b>9 / 9</b><br><b>Entities Extracted</b><br>100% success rate | <b>867K+</b><br><b>Rows Loaded to Warehouse</b><br>360K order items, 355K inventory | <b>20</b><br><b>dbt Models Built</b><br>9 staging + 11 marts | <b>77 / 78</b><br><b>dbt Tests Passed</b><br>1 warning (inventory stock) |
|----------------------------------------------------------------|-------------------------------------------------------------------------------------|--------------------------------------------------------------|--------------------------------------------------------------------------|

The pipeline ran successfully end-to-end, completing a full daily run in approximately 7 minutes. All critical data quality tests passed. A single warning was raised for negative inventory stock levels — a known data quality issue in the source system flagged intentionally by a custom dbt test.

# 2. Pipeline Architecture

The pipeline follows a modern ELT (Extract → Load → Transform) architecture with two separate PostgreSQL databases and a clear separation of concerns across each layer.

## 2.1 Infrastructure Overview

| Component    | Technology           | Purpose                                                    |
|--------------|----------------------|------------------------------------------------------------|
| lake_db      | PostgreSQL 15        | Raw ERP data landing zone (port 5433)                      |
| warehouse_db | PostgreSQL 15        | Transformed analytics warehouse (port 5434)                |
| Extractor    | Python + requests    | Pulls 9 entities from ERP API incrementally                |
| dlt Pipeline | dlt[postgres]        | Moves data from lake_db to warehouse_db with type coercion |
| dbt          | dbt-core 1.7.17      | Staging views, SCD2 snapshots, dimension & fact tables     |
| Airflow      | Apache Airflow 2.9.3 | Orchestrates the full pipeline on a daily schedule         |
| Docker       | Docker Compose       | Containerises all 7 services for reproducible execution    |

## 2.2 DAG Task Order

The Airflow DAG enforces strict task dependencies — each step only runs if the previous one succeeded:

| # | Task           | Description                                                                                                               |
|---|----------------|---------------------------------------------------------------------------------------------------------------------------|
| 1 | <b>extract</b> | Calls ERP REST API for all 9 entities using watermark-based incremental loading. Upserts records into lake_db raw schema. |

|   |                     |                                                                                                                  |
|---|---------------------|------------------------------------------------------------------------------------------------------------------|
| 2 | <b>dlt_load</b>     | Reads from lake_db, applies type coercion (e.g. string prices to numeric), upserts into warehouse_db raw schema. |
| 3 | <b>dbt_snapshot</b> | Runs SCD Type 2 snapshots for customers and products — captures history when segments or prices change.          |
| 4 | <b>dbt_staging</b>  | Builds 9 staging views: casts types, renames columns, flags soft-deletes and anomalous payments.                 |
| 5 | <b>dbt_marts</b>    | Builds all dimension and fact tables: 6 dims, 4 facts, and flagged_payments.                                     |
| 6 | <b>dbt_test</b>     | Runs 78 data quality tests across all models.                                                                    |

## 3. Kimball Dimensional Model

The warehouse follows Kimball star schema methodology. Facts are stored at the lowest grain, surrounded by conformed dimensions that can be joined across multiple fact tables.

### 3.1 Dimension Tables

| Dimension          | Rows  | Key Attributes                                                                       |
|--------------------|-------|--------------------------------------------------------------------------------------|
| dim_customer       | 5,000 | customer_key (SK), customer_id, segment, tier, city, state — SCD2 via snap_customers |
| dim_product        | 2,000 | product_key (SK), product_id, product_name, category, price — SCD2 via snap_products |
| dim_store          | 4     | store_key (SK), store_id, store_name, city, state                                    |
| dim_employee       | 50    | employee_key (SK), employee_id, full_name, role, store_key (FK)                      |
| dim_date           | 1,461 | date_key (SK), full_date, year, month, quarter, day_of_week                          |
| dim_payment_method | 5     | payment_method_key (SK), method_name                                                 |

### 3.2 Fact Tables

| Fact Table          | Rows    | Grain & Key Measures                                                                                     |
|---------------------|---------|----------------------------------------------------------------------------------------------------------|
| fct_sales           | 360,463 | One row per order line item. Measures: quantity, unit_price, gross_amount, discount_amount, net_revenue. |
| fct_payments        | 70,418  | One row per payment transaction. Measures: amount_paid, is_refund. Excludes flagged payments.            |
| fct_order_lifecycle | 80,000  | One row per order. Measures: days_to_deliver, current_status, order dates.                               |
| fct_inventory_daily | 344,073 | One row per product-store-day. Measures: quantity_in, quantity_out, closing_stock.                       |

## 4. Pipeline Execution Results

### 4.1 Extraction Layer

The Python extractor successfully pulled all 9 entities from the ERP REST API using watermark-based incremental loading. Records are upserted into the lake\_db raw schema on every run, ensuring idempotency.

| Entity              | Records Loaded | Notes                                                  |
|---------------------|----------------|--------------------------------------------------------|
| stores              | 4              | All 4 RetailCo store locations                         |
| employees           | 50             | Full staff roster across all stores                    |
| payment_methods     | 5              | Card, Mobile Money, Bank Transfer, Cash, USSD          |
| customers           | 5,000          | Paginated 4-page fetch; incremental on subsequent runs |
| products            | 2,000          | Full product catalogue                                 |
| orders              | 80,000         | Transaction orders with status tracking                |
| order_items         | 360,463        | Line-item detail for every order                       |
| payments            | 71,900         | Payment records including refunds and flagged items    |
| inventory_movements | 355,102        | Inbound/outbound stock movements                       |

Extraction complete in 18.9 seconds. Succeeded: 9/9 entities. All entities extracted successfully.

### 4.2 dbt Build Summary

dbt built 20 models (9 staging views + 11 mart tables) in 12.25 seconds. All 2 snapshots completed successfully capturing SCD Type 2 history for customers and products.

|                                          |                                               |                                                           |                                                    |
|------------------------------------------|-----------------------------------------------|-----------------------------------------------------------|----------------------------------------------------|
| <b>9</b><br>Staging Views<br>CREATE VIEW | <b>11</b><br>Mart Tables<br>SELECT into table | <b>2</b><br>Snapshots<br>snap_customers,<br>snap_products | <b>12.3s</b><br>Build Time<br>4 concurrent threads |
|------------------------------------------|-----------------------------------------------|-----------------------------------------------------------|----------------------------------------------------|

### 4.3 Data Quality — dbt Test Results

78 data quality tests were executed covering uniqueness, not-null constraints, referential integrity, accepted values, and 4 custom business logic tests.

| Test Category               | Count | Result                                   |
|-----------------------------|-------|------------------------------------------|
| not_null constraints        | 36    | PASS — No null primary or foreign keys   |
| unique constraints          | 19    | PASS — All surrogate keys are unique     |
| relationship / FK integrity | 9     | PASS — All fact-to-dim joins valid       |
| accepted_values             | 3     | PASS — Status codes and directions valid |

|                                                    |   |                                                             |
|----------------------------------------------------|---|-------------------------------------------------------------|
| Custom:<br>assert_fct_sales_positive_revenue       | 1 | PASS — No negative revenue in fct_sales                     |
| Custom:<br>assert_fct_payments_no_flagged          | 1 | PASS — Flagged payments excluded from fct_payments          |
| Custom:<br>assert_fct_order_lifecycle_one_row      | 1 | PASS — One row per order enforced                           |
| Custom:<br>assert_fct_inventory_non_negative_stock | 1 | WARN — 194,358 rows with negative stock (source data issue) |

Note: The inventory negative stock warning is an expected outcome. The source ERP system records outbound movements before inbound replenishments are processed, creating temporary negative stock positions. The custom test is configured to WARN (not fail) to surface this for analyst awareness.

# Business Insights from the Warehouse

The following analytical queries were run against the fully built warehouse to validate the data and surface business intelligence insights.

## Revenue by Store

All four RetailCo stores generated broadly comparable revenue, indicating consistent performance across locations. Abuja leads slightly by total revenue.

| Store                  | Total Revenue (₦)  | Total Orders |
|------------------------|--------------------|--------------|
| RetailCo Abuja         | ₦16,750,974,352.77 | 20,213       |
| RetailCo Kano          | ₦16,672,239,027.83 | 20,176       |
| RetailCo Port Harcourt | ₦16,336,276,995.32 | 19,963       |
| RetailCo Lagos         | ₦16,262,331,746.16 | 19,648       |

Combined total revenue across all stores: ₦66,021,822,122.27 from 80,000 orders.

## Top 10 Products by Revenue

Electronics and Home & Garden categories dominate the top revenue-generating products. Notably, gloves appear multiple times across categories, suggesting strong demand for this product type.

| #  | Product Name             | Category        | Revenue (₦)     | Units Sold |
|----|--------------------------|-----------------|-----------------|------------|
| 1  | Electronic Wooden Gloves | Home & Garden   | ₦115,032,912.80 | 1,944      |
| 2  | Tasty Steel Gloves       | Food & Beverage | ₦109,105,104.88 | 1,946      |
| 3  | Rustic Soft Pizza        | Home & Garden   | ₦107,186,085.80 | 1,911      |
| 4  | Electronic Metal Hat     | Sports          | ₦106,233,835.46 | 1,918      |
| 5  | Recycled Bronze Pants    | Clothing        | ₦103,144,794.79 | 2,050      |
| 6  | Fantastic Soft Gloves    | Books & Media   | ₦102,187,541.88 | 1,910      |
| 7  | Unbranded Fresh Sausages | Clothing        | ₦100,928,824.14 | 1,991      |
| 8  | Luxurious Cotton Hat     | Sports          | ₦98,174,576.85  | 1,868      |
| 9  | Licensed Granite Mouse   | Sports          | ₦96,861,325.11  | 2,171      |
| 10 | Bespoke Bronze Computer  | Automotive      | ₦94,735,357.53  | 2,034      |

## Payment Method Analysis

Card payments lead by total transaction value, but all five payment methods show remarkably even distribution — suggesting RetailCo's customers are broadly served across payment preferences with no single method dominating.

| Payment Method | Transaction Count | Total Amount (₦)   | Refund Count |
|----------------|-------------------|--------------------|--------------|
| Card           | 14,174            | ₦11,699,252,810.72 | 273          |

|               |        |                    |     |
|---------------|--------|--------------------|-----|
| Mobile Money  | 14,150 | ₦11,480,348,186.25 | 296 |
| Bank Transfer | 14,037 | ₦11,480,083,813.11 | 283 |
| Cash          | 14,027 | ₦11,361,056,689.89 | 284 |
| USSD          | 14,030 | ₦11,318,604,089.07 | 283 |

### Flagged Payments

The pipeline identifies and isolates anomalous payments into a dedicated flagged\_payments table, keeping fct\_payments clean for analysis.

| Flag Reason                  | Count | Total Amount                                             |
|------------------------------|-------|----------------------------------------------------------|
| flagged_zero                 | 734   | ₦0.00 — Zero-value payments requiring investigation      |
| flagged_unexplained_negative | 748   | -₦1,979,612.00 — Negative payments not tagged as refunds |

1,482 flagged payments (734 zero-value + 748 unexplained negatives) represent approximately 2.1% of total payment volume. These are isolated from the main fct\_payments table to ensure analytical accuracy.

### Order Lifecycle & Delivery Performance

The fct\_order\_lifecycle table tracks each order from placement to its current status, including delivery duration.

| Status                               | Order Count | Avg. Days to Deliver | % of Total |
|--------------------------------------|-------------|----------------------|------------|
| delivered                            | 48,078      | 13.0 days            | 60.1%      |
| pending / paid / shipped / cancelled | 31,922      | N/A                  | 39.9%      |

Average delivery time of 13.0 days across 48,078 delivered orders. 39.9% of orders are in transit or pending fulfilment.

## 6. Airflow Orchestration

The pipeline is fully orchestrated by Apache Airflow 2.9.3 running in a Docker container. The DAG is configured for daily execution with incremental data loading, exponential backoff retries, and a maximum of one concurrent run to prevent overlapping executions.

### 6.1 DAG Configuration

| Parameter       | Value                                                            |
|-----------------|------------------------------------------------------------------|
| dag_id          | retailco_pipeline                                                |
| Schedule        | @daily (00:00 UTC)                                               |
| Start Date      | 2025-01-01 (backfill-capable)                                    |
| Max Active Runs | 1 (prevents concurrent overlaps)                                 |
| Retries         | 2 per task, with exponential backoff                             |
| Max Retry Delay | 30 minutes                                                       |
| Catchup         | True — historical backfill supported                             |
| Operators Used  | PythonOperator (extract, dlt_load), BashOperator (all dbt tasks) |

### 6.2 Successful Run History

The pipeline achieved two consecutive fully successful end-to-end runs, confirming stability and idempotency of the incremental loading logic.

| Run ID                     | Start    | End      | Result                      |
|----------------------------|----------|----------|-----------------------------|
| manual_2026-05-31T20:38:45 | 20:57:23 | 21:20:54 | SUCCESS — All 6 tasks green |
| manual_2026-05-31T20:40:08 | 21:20:54 | 21:27:11 | SUCCESS — All 6 tasks green |

## 7. Conclusion

The RetailCo data pipeline successfully delivers a production-grade, end-to-end modern data engineering solution. Key achievements include:

- Full ELT pipeline from live REST API to analytics-ready star schema warehouse
- 867,000+ rows processed across 9 entities with incremental, idempotent loading
- Kimball dimensional model with 6 conformed dimensions, 4 fact tables, and SCD Type 2 history
- 78 automated data quality tests with 77 passing and 1 intentional warning surfacing a source data issue
- Daily Airflow orchestration with retry logic, backfill capability, and full observability via the UI
- Complete containerisation — the entire stack spins up with a single docker-compose up -d command
- Business insights surfaced: N66B+ total revenue, 13-day average delivery, 2.1% flagged payment rate

The pipeline is production-ready, well-tested, and documented. It demonstrates proficiency in the full modern data stack: ingestion, loading, transformation, orchestration, and data quality — all within a containerised environment.